

YONGHUI DONG



EDUCATION

Oct.2010

Jun.2014

● **Trento University, Italy**

Fondazione Edmund Mach, Italy

PhD in Analytical Chemistry

Supervisors: Dr.Pietro Franceschi, Dr.Fulvio Mattivi & Prof.Guella Graziano

Thesis: Mass Spectrometry Imaging: Looking Fruits at Molecular Level

Jun.2012

Dec.2012

● **Max Planck Institute for Chemical Ecology, Germany**

Visiting PhD student

Supervisor: Dr.Ales Svatos

Project: Mass spectrometry imaging of surface lipids on intact *Drosophila melanogaster* flies

Sep.2008

Sep.2010

● **Bologna University, Italy**

Munich University of Technology, Germany

BOKU University, Austria

Triple MSc. in Horticulture Science

Supervisors: Dr. Stefano Tartarini, Dr. Luca Dondini & Prof. Dieter Treutter

Thesis: QTL Analysis of Sugars and Organic Acids in Apricot

Sep.2003

Jun.2006

● **Northwest A&F University, China**

BSc. in Bioengineering

Supervisor: Prof. Zhihui Cheng

Thesis: Effects of UV-C on the control of cucumber powdery mildew



RESEARCH EXPERIENCE

Nay.

2024

Now

● **Professor**

Fujian Agriculture and Forestry University, China

CONTACT

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📄 [ResearchGate](#)

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RESEARCH INTERESTS

➤ Mass Spectrometry Imaging

➤ Metabolomics

➤ Stable isotope labeling

➤ Bioinformatics

SKILLS

Programming:

➤ R (9/10)

➤ Python (7/10)

➤ HTML/CSS (7/10)

➤ Git/GitHub (8/10)

➤ Docker (7/10)

Last updated on 2026-07-13.

- June. 2022
 |
 April. 2024
 ● **Researcher**
 Mass Spectrometry Imaging Unit, Weizmann Institute of Science, Israel
- Jan. 2021
 |
 May. 2022
 ● **Researcher**
 Blavatnik Center for Drug Discovery, Tel Aviv University, Israel
- Dec. 2014
 |
 Dec. 2020
 ● **Postdoctoral Research Fellow**
 Weizmann Institute of Science, Israel
 Supervisor: Prof. Asaph Aharoni
- Jul. 2014
 |
 Nov. 2014
 ● **Research Associate**
 Trento University, Italy
 Supervisor: Prof. Guella Graziano

PUBLICATIONS

- 2026
 ● **40. Complete biosynthesis of psychedelic tryptamines from three kingdoms in plants**
 Science Advances 12 (14), eaeb3034
 Paula Berman, Janka Höfer, Herschel Mehlman, Efrat Almekias-Siegl, Olga Khersonsky, [Yonghui Dong](#), Uwe Heinig, Liron Sulimani, Let Kho Hao, Shahar Cohen, Yoav Peleg, Sagit Meir, Ilana Rogachev, David Meiri, Sarel J Fleishman, Asaph Aharoni
- **39. Triple labeling of metabolites for metabolome analysis (TLEMMA): a stable isotope labeling approach for metabolite identification and network reconstruction**
 The Plant Journal 123 (1), e70333
[Yonghui Dong](#), Liron Feldberg, Yi Wang, Uwe Heinig, Ilana Rogachev, and Asaph Aharoni.
- **38. AlphaFold-Guided Bespoke Gene Editing Enhances Field-Grown Soybean Oil Contents**
 Advanced Science 12 (23), 2500290
 Jie Wang, Li Zhang, Shoudong Wang, Xin Wang, Suning Li, Pingping Gong, Mengyan Bai, Arnav Paul, Nathan Tvedt, Hengrui Ren, Maoxiang Yang, Zhihui Zhang, Shaodong Zhou, Jiayi Sun, Xianjin Wu, Huaqin Kuang, Zhenghua Du, [Yonghui Dong](#), Xiaolei Shi, Meina Li, Diwakar Shukla, Long Yan, Yuefeng Guan
- **37. Triterpenoid saponins in tea plants: A spatial and metabolic analysis using UPLC-QTOFMS, molecular networking, and DESI-MSI**
 Food Chemistry 475, 143323
 Du, Zhenghua, Ying Zhou, Shuang Guo, [Yonghui Dong](#), Yongquan Xu, and Xiaomin Yu.
- 2025
 ● **36. Genetic control over biogenic crystal morphogenesis in zebrafish**
 Nature Chemical Biology 21 (3), 383-392
 Deis, R., Lerer-Goldshtein, T., Baiko, O., Eyal, Z., Brenman-Begin, D., Goldsmith, M., Kaufmann, S., Heinig, U., [Yonghui Dong](#), Lushchekina, S. and Varsano, N.,

- **35. A cellulose synthase-like protein governs the biosynthesis of Solanum alkaloids**
 Science 386 (6728), eadq5721
 Adam Jozwiak, Sayantan Panda, Ryota Akiyama, Ayano Yoneda, Naoyuki Umemoto, Kazuki Saito, Shuhei Yasumoto, Toshiya Muranaka, Sachin A Gharat, Yana Kazachkova, [Yonghui Dong](#), Shlomy Arava, Inna Goliand, Reinat Nevo, Ilana Rogachev, Sagit Meir, Masaharu Mizutani, Asaph Aharoni
- **34. Metabolite Annotation through Stable Isotope Labeling**
 TrAC Trends in Analytical Chemistry 181, 118037
[Yonghui Dong](#), Liron Feldberg, Asaph Aharoni, and Uwe Heinig
- **33. A Comprehensive Spatially Resolved Metabolomics Dataset for Lampreys**
 Scientific Data 11 (1), 1075
 Gou, Meng, Xiaxia Wang, Xuyuan Duan, Yaocen Wang, Yue Pang, [Yonghui Dong](#) (corresponding author)
- **32. The biosynthetic pathway of the hallucinogen mescaline and its heterologous reconstruction**
 Molecular Plant 17 (7), 1129-1150
 Paula Berman, Luis Alejandro de Haro, Ana-Rita Cavaco, Sayantan Panda, [Yonghui Dong](#), Nikolay Kuzmich, Gabriel Lichtenstein, Yoav Peleg, Hila Harat, Adam Jozwiak, Jianghua Cai, Uwe Heinig, Sagit Meir, Ilana Rogachev, Asaph Aharoni
- **31. New wine in an old bottle: ABCB19, known as auxin exporter, also exports brassinosteroids**
 The Innovation Life 2 (2), 100072-1-100072-2
[Yonghui Dong](#), Ning Su, and Yuqin Zhang
- **30. MetaboReport: from metabolomics data analysis to comprehensive reporting**
 Bioinformatics 40 (6), btac373
[Yonghui Dong](#), and Sergey Malitsky.
- **29. Spatial Metabolomics Reveals the Multifaceted Nature of Lamprey Buccal Gland and Its Diverse Mechanisms for Blood-Feeding**
 Communications Biology 6 (1), 881
 Meng Gou, Xuyuan Duan, Jun Li, Yaocen Wang, Qingwei Li, Yue Pang, [Yonghui Dong](#) (corresponding author)
- **28. A complete prostaglandin pathway from synthesis to inactivation in the oral gland of the jawless vertebrate lamprey, Lethenteron camtschaticum**
 Developmental & Comparative Immunology 148, 104903
 Wang, Yaocen, [Yonghui Dong](#) (co-first author), Xuyuan Duan, Yimu Luan, Qingwei Li, Yue Pang, Feng Sun, and Meng Gou
- **27. Parallel evolution of cannabinoid biosynthesis**
 Nature Plants 9 (5), 817-831
 Paula Berman, Luis Alejandro de Haro, Adam Jozwiak, Sayantan Panda, Zoe Pinkas, [Yonghui Dong](#), Jelena Cveticanin, Ranjit Barbole, Rotem Livne, Tali Scherf, Eyal Shimoni, Smadar Levin-Zaidman, Nili Dezarella, Ekaterina Petrovich-Kopitman, Sagit Meir, Ilana Rogachev, Prashant D Sonawane, Asaph Aharoni

2023

- **26. PICA: Pixel Intensity Correlation Analysis for Deconvolution and Metabolite Identification in Mass Spectrometry Imaging**
Analytical Chemistry 95 (2), 1652-1662
[Yonghui Dong](#), Nir Shachaf, Liron Feldberg, Ilana Rogachev, Uwe Heinig, Asaph Aharoni
- **25. Image to insight: exploring natural products through mass spectrometry imaging**
Natural Product Reports 39 (7), 1510-1530
[Yonghui Dong](#), Asaph Aharoni
- **24. TMT-based quantitative proteomics reveals protein biomarkers from cultured Pacific abalone (*Haliotis discus hannai*) in different regions**
Food Chemistry: X 14, 100355
YimuLuan, [Yonghui Dong](#) (co-first author), Xuyuan Duan, Xiuli Wang, Yue Pang, Qingwei Li, MengGou
- **23. The metabolic and proteomic repertoires of periderm tissue in skin of the reticulated Sikkim cucumber fruit**
Horticulture Research 9, uhac092
Gulab Chand Arya, [Yonghui Dong](#) (co-first author), Uwe Heinig, Nir Shahaf, Yana Kazachkova, Elinor Aviv-Sharon, Gal Nomborg, Ofir Marinov, Ekaterina Manasheva, Asaph Aharoni, Hagai Cohen
- **22. Quantitative Trait Loci Mapping and Identification of Candidate Genes Linked to Fruit Acidity in Apricot (*Prunus armeniaca* L.)**
Frontiers in Plant Science 13, 838370
Luca Dondini, Cecilia Domenichini, [Yonghui Dong](#), Fabio Gennari, Daniele Bassi, Stefano Foschi, Martina Lama, Marco Adami, Paolo De Franceschi, Claudia Cervellati, Lorenzo Bergonzoni, Sara Alessandri, Stefano Tartarini
- **21. Saliva metabolome alterations after acute stress**
Scientific Reports 12 (1), 18470
Liat Morgan, Rune Isak Dupont Birkler, Shira Shaham-Niv, [Yonghui Dong](#), Tal Wachsman, Lior Carmi, Boris Yakobson, Lihi Adler-Abramovich, Hagit Cohen, Joseph Zohar, Melissa Bateson, Ehud Gazit

2022

- **20. RawHummus: an R Shiny App for Automated Raw Data Quality Control in Metabolomics**
Bioinformatics 38 (7), 2072-2074
[Yonghui Dong](#), Yana Kazachkova, Meng Gou, Liat Morgan, Tal Wachsman, Ehud Gazit, Rune Isak Dupont Birkler
- **19. CCWeights: an R package and web application for automated evaluation and selection of weighting factors for accurate quantification using linear calibration curve**
Bioinformatics Advances 1 (1), vbab029
[Yonghui Dong](#), Tal Wachsman, Liat Morgan, Ehud Gazit, Rune Isak Dupont Birkler
- **18. Characterization of the PRODUCTION OF ANTHOCYANIN PIGMENT 1 Arabidopsis Dominant Mutant using DLEMMA Dual Isotope Labeling Approach**
Phytochemistry 186, 112740
[Yonghui Dong](#), Liron Feldberg, Ilana Rogachev, Asaph Aharoni

- 2021 ● **17. The GORKY glycoalkaloid transporter is indispensable for preventing tomato bitterness**
Nature Plants 7 (4), 468-480
Yana Kazachkova, Itay Zemach, Sayantan Panda, Samuel Bocobza, Andrii Vainer, Ilana Rogachev, [Yonghui Dong](#), Shifra Ben-Dor, Dorottya Veres, Christa Kanstrup, Sophie Konstanze Lambertz, Christoph Crocoll, Yangjie Hu, Eilon Shani, Simon Michaeli, Hussam Hassan Nour-Eldin, Dani Zamir, Asaph Aharoni
- **16. High mass resolution, spatial metabolite mapping enhances the current plant gene and pathway discovery toolbox**
New Phytologist 228 (6), 1986-2002
[Yonghui Dong](#), Prashant Sonawane, Hagai Cohen, Guy Polturak, Liron Feldberg, Shelly Hen Avivi, Ilana Rogachev, Asaph Aharoni
- 2020 ● **15. Rhizosphere microbiome mediates systemic root metabolite exudation by root-to-root signaling**
Proceedings of the National Academy of Sciences 117 (7), 3874-3883
Elisa Korenblum, [Yonghui Dong](#), Jędrzej Szymanski, Sayantan Panda, Adam Jozwiak, Hassan Massalha, Sagit Meir, Ilana Rogachev, Asaph Aharoni
- **14. Miso: an R package for multiple isotope labeling assisted metabolomics data analysis**
Bioinformatics 35 (18), 3524-3526
[Yonghui Dong](#), Liron Feldberg, Asaph Aharoni
- **13. A Multilevel Study of Melon Fruit Reticulation Provides Insight into Skin Lignosubersization Hallmarks**
Plant Physiology 179 (4), 1486-1501
Hagai Cohen, [Yonghui Dong](#), Jędrzej Szymanski, Justin Lashbrooke, Sagit Meir, Efrat Almekias-Siegl, Viktoria Valeska Zeisler-Diehl, Lukas Schreiber, Asaph Aharoni
- 2019 ● **12. In plaque-mass spectrometry imaging of a bloom-forming alga during viral infection reveals a metabolic shift towards odd-chain fatty acid lipids**
Nature Microbiology 4 (3), 527-538
Guy Schleyer, Nir Shahaf, Carmit Ziv, [Yonghui Dong](#), Roy A Meoded, Eric JN Helfrich, Daniella Schatz, Shilo Rosenwasser, Ilana Rogachev, Asaph Aharoni, Jörn Piel, Assaf Vardi
- **11. DLEMMA-MS-imaging for identification of spatially localized metabolites and metabolic network map reconstruction**
Analytical Chemistry 90 (17), 10231-10238
Liron Feldberg, [Yonghui Dong](#) (co-first author), Uwe Heinig, Ilana Rogachev, Asaph Aharoni
- 2018 ● **10. Mapping of cell wall aromatic moieties and their effect on hygroscopic movement in the awns of stork's bill**
Cellulose 25 (7), 3827-3841
Yael Abraham, [Yonghui Dong](#), Asaph Aharoni, Rivka Elbaum
- **09. TLC surface integrity affects the detection of alkali adduct ions in TLC-MALDI analysis**
Analytical and Bioanalytical Chemistry 409 (24), 5661-5666
[Yonghui Dong](#), Ruggero Ferrazza, Andrea Anesi, Graziano Guella, Pietro Franceschi

- 2017
- **08. Engineered gray mold resistance, antioxidant capacity, and pigmentation in betalain-producing crops and ornamentals**
Proceedings of the National Academy of Sciences 114 (34), 9062-9067
Guy Polturak, Noam Grossman, David Vela-Corcia, [Yonghui Dong](#), Adi Nudel, Margarita Pliner, Maggie Levy, Ilana Rogachev, Asaph Aharoni
 - **07. Impact of tissue surface properties on the desorption electrospray ionization imaging of organic acids in grapevine stem**
Rapid Communications in Mass Spectrometry 30 (6), 711-718
[Yonghui Dong](#), Graziano Guella, Pietro Franceschi
 - **06. More than pictures: when MS imaging meets histology**
Trends in Plant Science 21 (8), 686-698
[Yonghui Dong](#), Bin Li, Asaph Aharoni
 - **05. Sample preparation for mass spectrometry imaging of plant tissues: a review**
Frontiers in Plant Science 7, 60
[Yonghui Dong](#), Bin Li, Sergey Malitsky, Ilana Rogachev, Asaph Aharoni, Filip Kaftan, Aleš Svatoš, Pietro Franceschi
 - **04. Identification of microRNAs and their targets associated with fruit-bagging and subsequent sunlight re-exposure in the “Granny Smith” apple exocarp using high-throughput sequencing**
Frontiers in Plant Science 7, 27
Dong Qu, Fei Yan, Rui Meng, Xiaobing Jiang, Huijuan Yang, Ziyi Gao, [Yonghui Dong](#), Yazhou Yang, Zhengyang Zhao
- 2016
- **03. Analytical capabilities of mass spectrometry imaging and its potential applications in food science**
Trends in Food Science & Technology 47, 50-63
Bin Li, Sage JB Dunham, [Yonghui Dong](#), Sohee Yoon, Maomao Zeng, Jonathan V Sweedler
- 2015
- **02. High production of small organic dicarboxylate dianions by DESI and ESI**
Journal of The American Society for Mass Spectrometry 26 (3), 386-389
[Yonghui Dong](#), Graziano Guella, Fulvio Mattivi, Pietro Franceschi
- 2012
- **01. Combining intensity correlation analysis and MALDI imaging to study the distribution of flavonols and dihydrochalcones in Golden Delicious apples**
Journal of Experimental Botany 63 (3), 1123-1133
Pietro Franceschi, [Yonghui Dong](#), Kerstin Strupat, Urska Vrhovsek, Fulvio Mattivi



POSTERS & TALKS

- 2025
- **ShinyCardinal: an Comprehensive Platform for Mass Spectrometry Imaging Data Analysis (Invited Speaker)**
The Second International Plant SpatioTemporal Omics Research Conference (STOC Plant 2025) (Beijing, China)
 - **The use of Stable Isotope Labeling in Plant Metabolomics (Invited Speaker)**
Frontiers in Biology Advances Forum (Chongqing, China)

- 2022 ● **How vampires suck blood? (Talk)**
First Israel Metabolomics Meeting (Israel)
- **Software solutions in untargeted and targeted metabolomics (Poster)**
First Israel Metabolomics Meeting (Israel)
- 2021 ● **Pixel-wise Colocalization Analysis for Metabolite Identification in Mass Spectrometry Imaging (Talk)**
Waterman Seminar Series on Bioinformatics (Germany)
- 2017 ● **Application of MS Imaging in Plant Sciences (Talk)**
Mass Spectrometry Imaging Symposium (France)
- **More than Pictures: When MS Imaging Meets Histology (Talk)**
Spring School for Advanced Imaging in Biological Research (Israel)
- 2015 ● **Mass Spectrometry Imaging of Plant Metabolites (Poster)**
Isranalytica 2015 (Israel)
- 2014 ● **Tissue Surface Properties Jeopardize Quantitative DESI Imaging of Organic Acids in Grapevine Stem (Poster)**
20th International Mass Spectrometry Conference (Switzerland)
- 2013 ● **MS Imaging of Metabolites in Fruits (Poster)**
3rd MS Food day (Italy)
- 2012 ● **MS Imaging of Small Metabolites in Fruits (Talk)**
Ourense Conference on Imaging Mass Spectrometry (Spain)